

Detection of Acid-Fast Bacilli in Slit-Skin Smear Microscopy Using Mask R-CNN Algorithm

Cabunilas Joshua John L., Ponio Kean Jayther P., Villaverde Jocelyn F. and Dofitas Belen Lardizabal

¹jjlcabunilas@mymail.mapua.edu.ph, ²kjpponio@mymail.mapua.edu.ph, ³jfvillaverde@mapua.edu.ph,

⁴belendofitas@gmail.com

1. Introduction

This study focuses on the development of a diagnostic system for acid-fast bacilli (AFB) detection in leprosy cases. AFB, known to be responsible for causing Leprosy, can be identified and quantified through slit skin smear examinations. To automate this process, we propose the implementation of a Mask R-CNN (Region-based Convolutional Neural Network) model with a RESNET50 backbone architecture. The Model generates three outputs: classification, bounding boxes, and instance segmentation, enabling the detection, classification, and quantification of AFB. To train the Model, a dataset of over 300 images of AFB was collected and manually annotated. Data augmentation techniques were applied to increase the dataset size. The Model was trained to classify AFB into three categories: solids, fragmented, and clumps. The proposed system provides a reliable and automated AFB detection and classification solution, enhancing Leprosy's diagnostic process.

2. Methodology

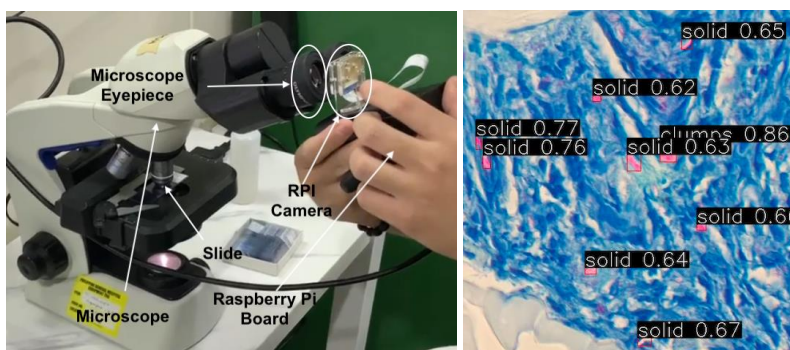


Fig. 1. Experimental Setup (LEFT), System Output (RIGHT)

The proposed device shown in figure 1 (LEFT) is designed to be portable and conveniently sized, making it ideal for field use and remote locations. The hardware setup includes a RPI Camera connected to the Raspberry Pi board. The camera is positioned at the eyepiece of the microscope, capturing images of the AFB samples. These images are then processed by an algorithm specifically developed for this purpose. The algorithm analyses the captured image, employing sophisticated image processing techniques to classify whether the AFB distribution is solid, clumped, fragmented or a combination of three or two. Furthermore, the algorithm can count the number of AFBs detected in the image, providing valuable information for diagnostic and research purposes. Overall, this portable device integrates advanced image recognition and processing technology with a Raspberry Pi-based hardware setup to enable efficient and accurate classification and quantification of acid-fast bacilli. Its potential applications range from field research and diagnostics to supporting healthcare professionals in remote areas where access to specialized laboratories and equipment may be limited.

Figure 1 (RIGHT) shows the image that was captured by the Raspberry Pi camera. This is the Actual image retrieved from the Philippine General Hospital (PGH). The classification clumps shown in the image are very small; thus, it does not capture fine detail due to the camera's quality and the object's size, but Bacilli were detected by the MASK R-CNN.

Table 1. Test Result Table

Test Images	Mask RCNN Classification	Actual Classification	Mask RCNN Quantification	Actual Quantification	Percent Difference
Image 1	Fragmented	Fragmented	1	1	0%
Image 2	Solids	Solids	10	13	28.57%
Image 3	Clumps	Clumps	3	13	26.087%
Image 4	Solids	Solids	7	7	0%
Image 5	Fragmented Solids	Fragmented Solids	2	2	0%
			10	14	33.33%
Image 6	Fragmented Solids	Fragmented Solids	4	6	40%
			2	3	40%
Image 7	Clumps	Clumps	5	3	50%
Image 8	Fragmented Solids	Fragmented Solids	1	1	0%
			1	1	0%
Image 9	Fragmented Solids	Fragmented Solids	4	4	0%
			8	8	0%
Image 10	Fragmented Solids	Fragmented Solids	2	1	66.67%
			6	6	0%
Image 11	Clumps	clumps	15	29	63.64%
Image 12	Fragmented Solids	Fragmented Solids	2	2	0%
			1	1	0%
Image 13	Clumps	Clumps	1	1	0%
	Fragmented Solids	Fragmented Solids	1	2	66.67%
			6	15	85.71%
Image 14	Fragmented Solids	Fragmented Solids	3	3	0%
			4	4	0%
Image 15	Clumps	Clumps	1	1	0%
Image 16	Solids	Solids	5	10	66.67%
Image 17	Clumps	Clumps	12	14	15.38%
Image 18	Solids	Solids	8	8	0%
Image 19	Solids	Solids	10	12	18.18%
Image 20	Fragmented Solids	Fragmented Solids	1	1	0%
			15	17	12.5%
Image 21	Solids	Solids	7	7	0%
Image 22	Solids	Solids	10	12	18.18%
Image 23	Fragmented Solids	Fragmented Solids	3	2	40%
			10	5	66.67%

Testing Datasets were used for comparing MASK R-CNN quantifications and actual quantifications. These datasets were not used in training the Model, and it also shows the percentage difference between the two quantifications. Based on the actual classification, MASK R-CNN performs well in classifying the Bacilli, whether clumps, solids, or fragments. However, in terms of quantifications, it cannot detect some very small and faded Bacilli. Images with a 0% difference between the actual and predicted are images containing very large and not faded Bacilli, which is distinguishable. To calculate for overall percent difference of the system, use equation 2, shown below. the overall percent difference of the system is 21.74%. Based on the overall difference of the system, the Model was able to classify and quantify the Bacilli. To improve the overall accuracy of the Model, more datasets that include clear images are needed.

$$\text{Overall Percent Difference} = \frac{\text{Sum of percent difference}}{\text{Total number of classification per image}} \quad (2)$$

3. Conclusion

The Model performs well in classifying the Bacilli based on the result table. Sometimes, the Model has difficulty distinguishing the solids and the fragmented Bacilli because they look similar. The factor that affects the Model from distinguishing the solids and fragmented is that some Bacilli show as faded and are classified as solids in actuality. Still, the Model predicts that faded Bacilli are considered fragmented. Since the Model was trained using a medium to large-looking Bacilli, factors that affect the accuracy of the Model are a test or actual images having a very small and complex-looking Bacilli example of this is if 3-5 Bacilli are stacked together, the Model is not able to detect. Another factor is the quality of the captured image. Since the camera used is a Raspberry version 1 camera, having 5 megapixels does not produce high-quality images, which loses the detail of a very small Bacilli. We conclude that a Raspberry Pi camera was used to capture Acid-Fast Bacilli, the MASK R-CNN algorithm was used for classifying and quantifying Acid-Fast Bacilli, and we were able to validate and quantify the output using percent difference, the overall percent difference based on the result is 21.74%.